Applied Regression Homework 2 225040015

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Conceptual Questions

Question 1

Table 1 reports p-values for each coefficient in the multiple regression of `sales` on `TV`, `radio`, and `newspaper` budgets. Each p-value tests the null hypothesis that the corresponding advertising channel has *no linear association with sales, holding the other two channels fixed*. Concretely:

- **TV:** H_0: TV budget has no association with sales (given radio and newspaper). p $< 0.0001 \Rightarrow$ strong evidence TV is associated with higher sales.
- **Radio:** H_0: radio budget has no association with sales (given TV and newspaper). p $< 0.0001 \Rightarrow$ strong evidence radio is associated with higher sales.
- **Newspaper:** H_0: newspaper budget has no association with sales (given TV and radio). p $= 0.8599$ $\Rightarrow$ no evidence of a linear association with sales once TV and radio are included.
- **Intercept:** H_0: expected sales is zero when all ad budgets are zero. p $< 0.0001 \Rightarrow$ baseline sales is significantly different from zero.

Question 5

Without an intercept, the fitted value is $\hat{y}_i = x_i b\hat{e}ta$ with

$$\hat{\beta} = \frac{\sum_{i'=1}^{n} x_{i'} y_{i'}}{\sum_{i'=1}^{n} x_{i'}^2}.$$

Thus

$$\hat{y}_i = x_i \cdot \frac{\sum_{i'=1}^{n} x_{i'} y_{i'}}{\sum_{i'=1}^{n} x_{i'}^2} = \sum_{i'=1}^{n} \left(\frac{x_i x_{i'}}{\sum_{j=1}^{n} x_j^2} \right) y_{i'}.$$

Therefore $a_{i'} =$
$dfracx_i x_{i'} sum_{j=1}^{n} x_j^2$.

Question 8: Simple Linear Regression

We fit the model

$$\text{mpg} = \beta_0 + \beta_1 \cdot \text{horsepower} + \varepsilon.$$

(a) Model summary and interpretation

- **Relationship:** The p-value for horsepower is 7.03×10^{-81}, indicating a statistically significant relationship.
- **Strength:** $R^2 = 0.6059$, so horsepower explains about 60.6% of the variability in mpg.
- **Direction:** The slope is negative ($\hat{\beta}_1 = -0.1578$), so higher horsepower is associated with lower mpg.
- **Prediction at horsepower = 98:** $\widehat{\text{mpg}} = 24.47$.
  - 95% confidence interval for mean response: [23.97, 24.96]
  - 95% prediction interval for a new observation: [14.81, 34.12]

(b) Scatter plot with regression line

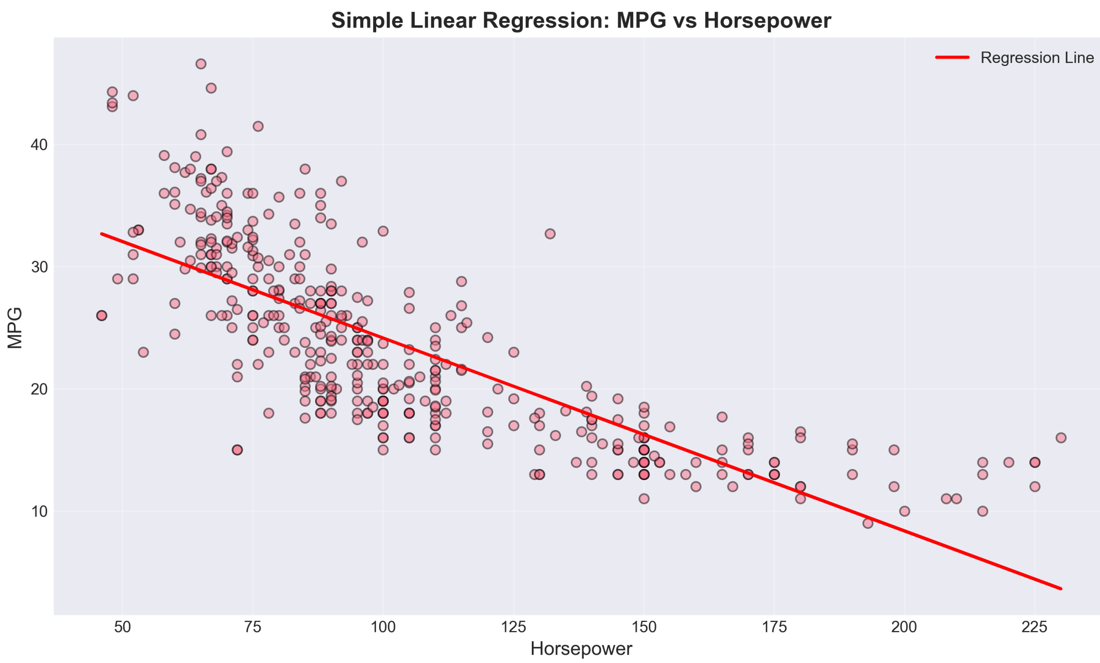

Figure 1: MPG vs. horsepower with least squares regression line.

(c) Diagnostic plots

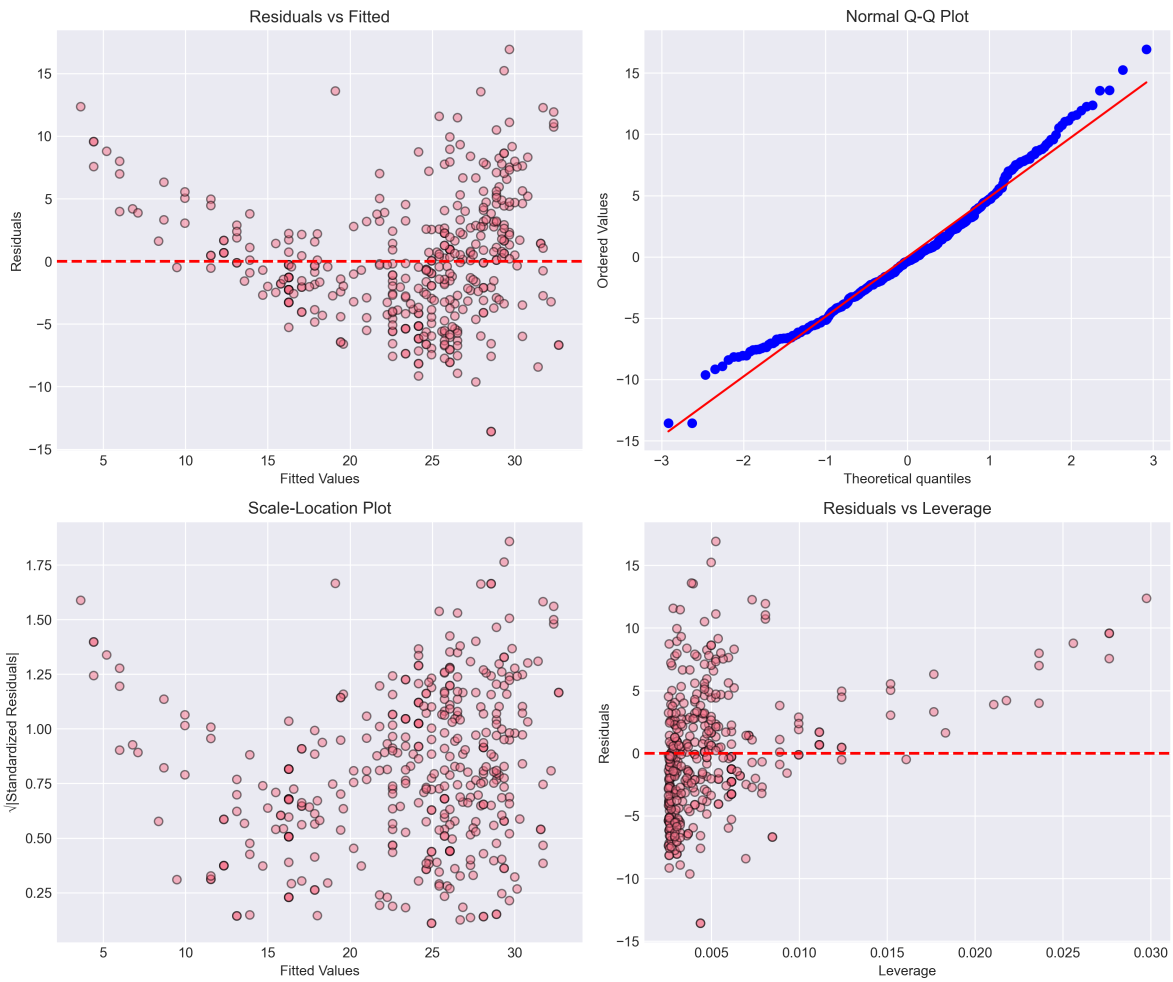

Figure 2: Diagnostic plots for the simple linear regression.

Comments: The residuals show mild patterns and increasing spread at higher fitted values, suggesting some nonlinearity and heteroscedasticity. The Q–Q plot is close to linear in the center but deviates in the tails, indicating slight non-normality.

Question 9: Multiple Linear Regression

We fit the model

$$\text{mpg} = \beta_0 + \beta_1\,\text{cylinders} + \beta_2\,\text{displacement} + \beta_3\,\text{horsepower} + \beta_4\,\text{weight} + \beta_5\,\text{acceleration} + \beta_6\,\text{year} + \beta_7\,\text{origin} + \varepsilon.$$

(a) Scatterplot matrix

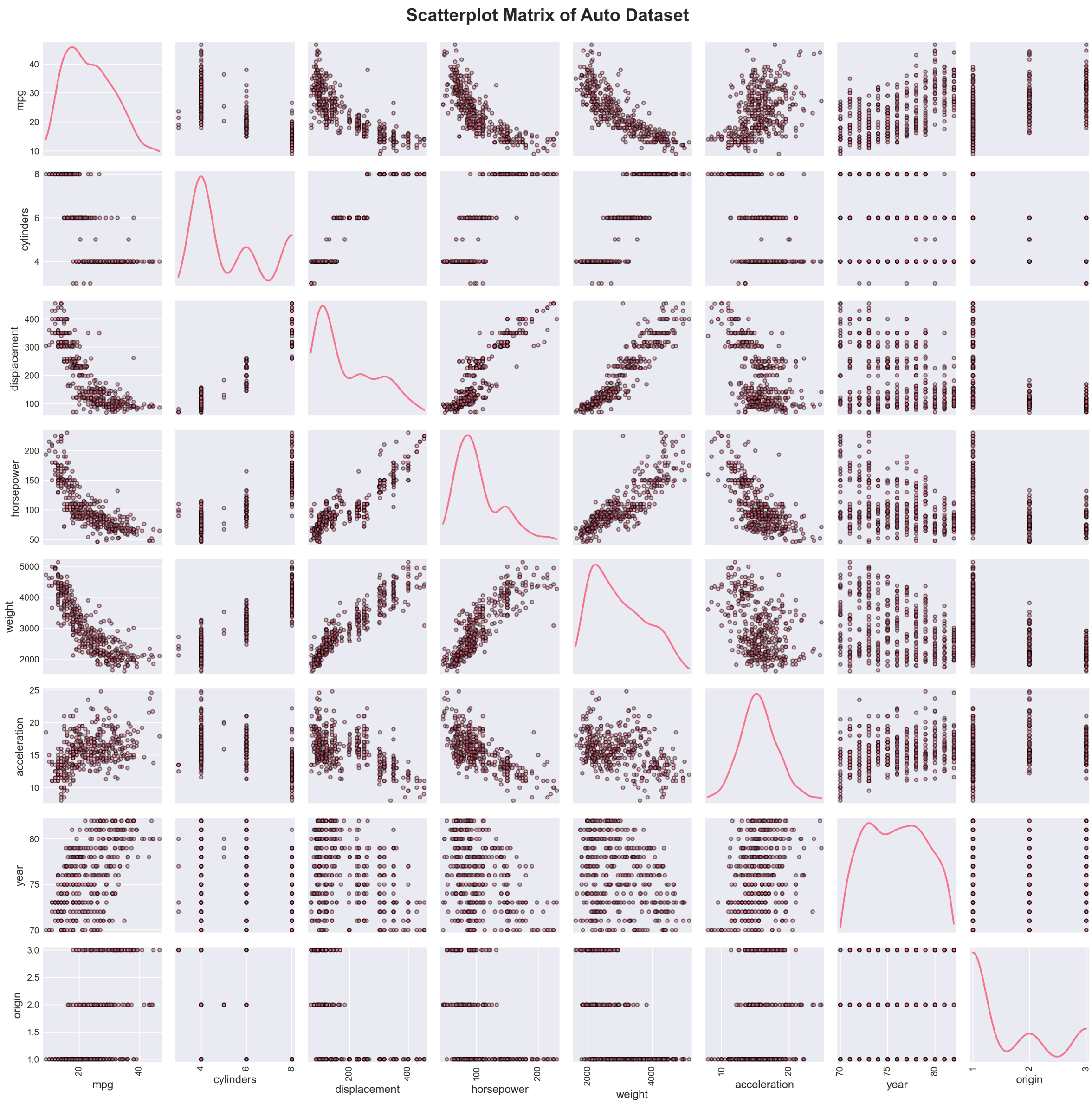

Figure 3: Scatterplot matrix of numeric variables in the Auto dataset.

(b) Correlation matrix

Strong negative correlations exist between mpg and displacement/weight/horsepower. Many predictors are also highly correlated with each other, indicating potential multicollinearity.

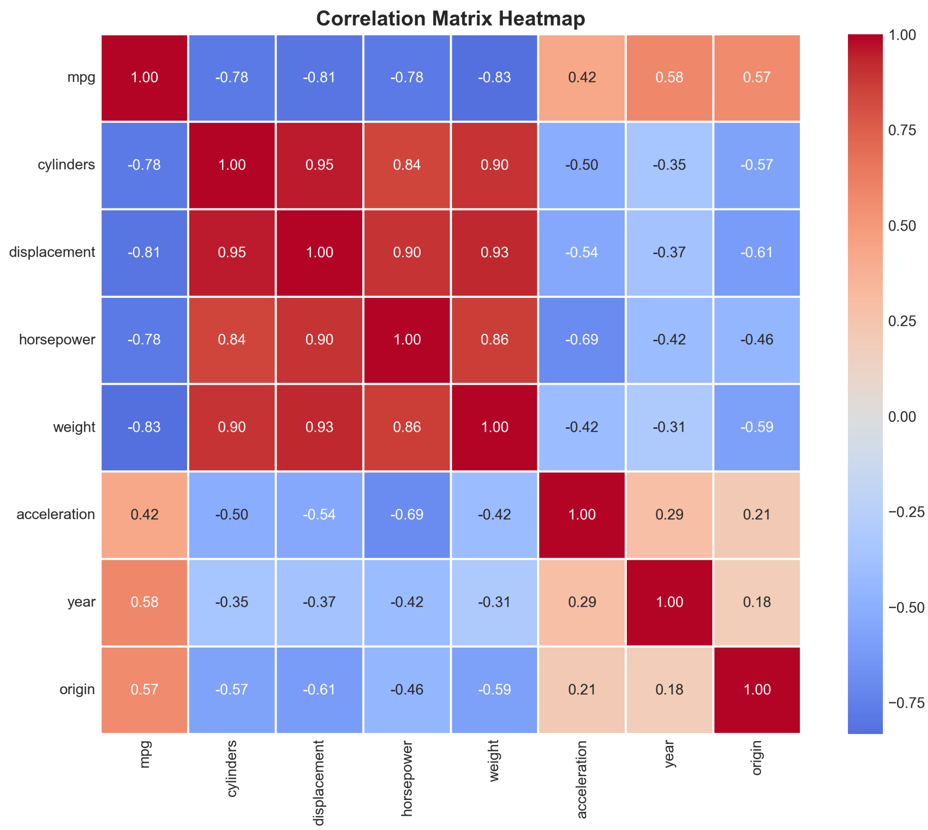

Figure 4: Correlation heatmap for numeric variables.

(c) Multiple regression summary

The overall model fit is strong: $R^2 = 0.8215$ and adjusted $R^2 = 0.8182$. Statistically significant predictors ($p < 0.05$) are:

- displacement ($p = 0.0084$)
- weight ($p < 0.0001$)
- year ($p < 0.0001$)
- origin ($p < 0.0001$)

The condition number is large (8.59×10^4), suggesting multicollinearity among predictors.

(d) Diagnostic plots and leverage

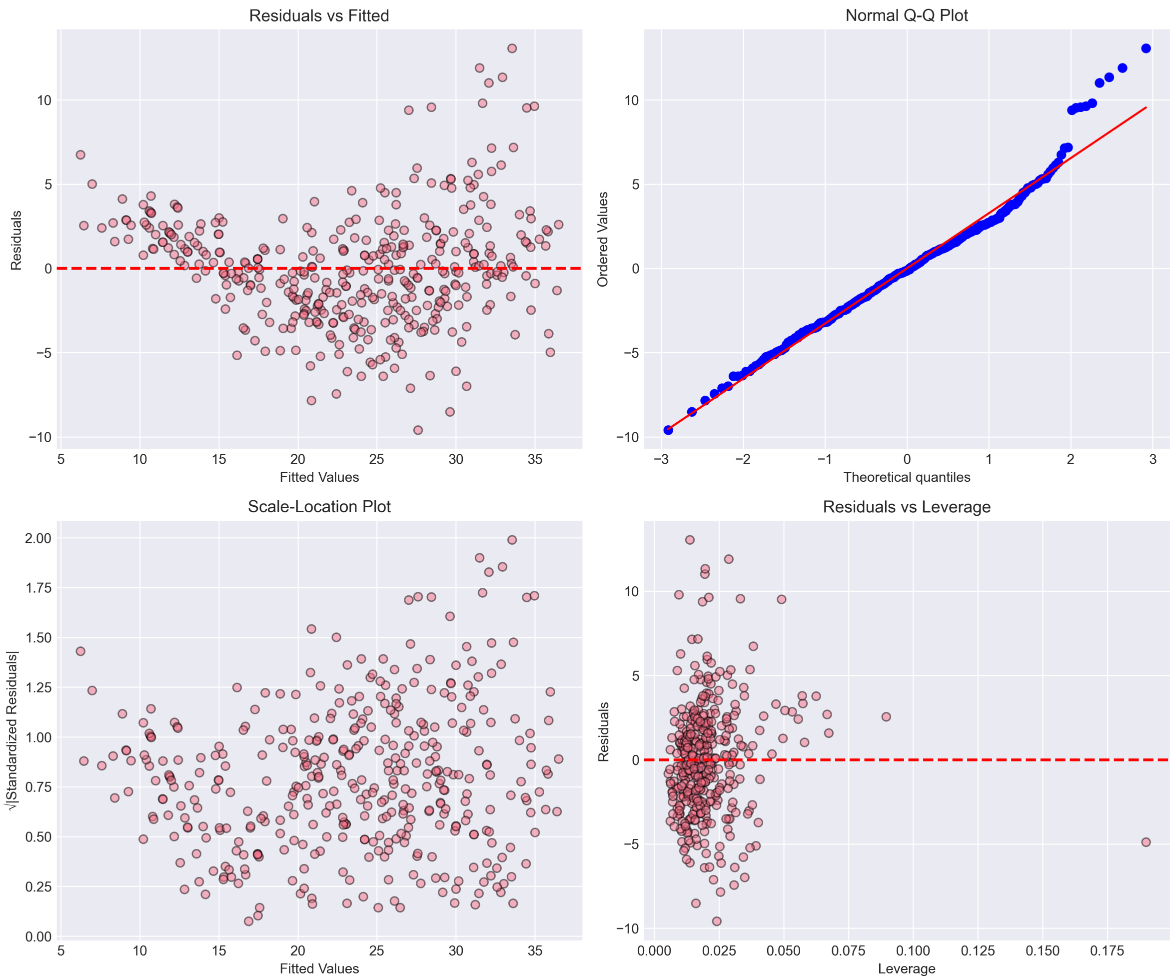

Figure 5: Diagnostic plots for the multiple linear regression.

High leverage points (leverage $> 3\times$ mean) were identified at indices: 8, 13, 26, 27, 28. The residual plots show mild heteroscedasticity and some tail deviations in the Q–Q plot.

(e) Interaction effects

Interaction models tested:

- displacement $\times$ weight: significant ($p < 0.0001$)
- horsepower $\times$ weight: significant ($p < 0.0001$)
- acceleration $\times$ year: not significant ($p = 0.5473$)

Thus, interactions involving weight appear important.

(f) Transformations

Several transformations were tested; the best improvement was obtained with log(weight).

- Original R^2: 0.8215

- log(weight) model R^2: 0.8459 (improvement of 0.0245)

This suggests a log transformation of weight better captures nonlinearity and improves fit.

Conclusion

Simple linear regression shows a strong negative relationship between horsepower and mpg. Multiple regression improves explanatory power, highlighting weight, year, origin, and displacement as key predictors. Diagnostics indicate mild heteroscedasticity and tail non-normality, while interaction and transformation analyses suggest that weight-related nonlinear effects are important.